

MIRA: Longitudinal Analysis

Dynamic and reproducible statistical report

2026-09-14

Table of contents

1	Abstract	2
2	Analysis context	2
2.1	Resolved configuration	2
2.1.1	Time vars	2
2.1.2	Time labels	2
2.1.3	Auto detected	2
2.1.4	Specified manually	2
2.1.5	Sources	3
2.1.6	Analyses	3
2.2	Dataset overview	3
2.2.1	Column profile	3
2.3	Detected variables	3
2.3.1	Id candidates	3
2.3.2	Longitudinal	3
2.3.3	Longitudinal map	3
3	Outcome: SCORE	4
3.1	Analysis settings	4
3.2	Overview	4
3.3	Descriptive statistics	4
3.4	Missing data	5
3.4.1	By time point	5
3.4.2	By participant	6
3.5	Within-participant changes	7
3.6	Variability and ICC	8
3.7	Individual trajectories	8
3.8	Advanced longitudinal tests	11
3.8.1	Estimated marginal means and contrasts	11
3.8.2	Repeated-measures ANOVA	11
3.8.3	Friedman test	11
3.9	Advanced longitudinal models	12
3.9.1	Random-slope mixed-effects model	12
3.9.2	NLME models	13
3.9.3	GEE models	13
3.10	Robust inference	13
3.10.1	Cluster-robust CR2 inference	13
3.11	Effect sizes	13
3.11.1	Paired cohens dz	13
3.11.2	Paired rank biserial	13
3.11.3	Friedman test	13
3.11.4	Mixed models	13
3.12	Multiplicity families	14

3.12.1 Friedman posthoc	14
3.13 Sensitivity analyses	14
3.14 Diagnostics and adaptations	15
3.14.1 Automatic adaptations	15
4 Descriptive summary	15

1 Abstract

This report presents the observed analytical results for score.

2 Analysis context

2.1 Resolved configuration

Id: subject_id

Id generated: FALSE

2.1.1 Time vars

Score: score__t0, score__t1, score__t2

2.1.2 Time labels

Score: t0, t1, t2

Improvement direction: unknown

Stable threshold: 0

Variable pattern: auto

2.1.3 Auto detected

Id: TRUE

Time vars: TRUE

Time labels: TRUE

Arm: TRUE

Reference arm: TRUE

Covariates: TRUE

Improvement direction: TRUE

Stable threshold: TRUE

2.1.4 Specified manually

Id: FALSE

Time vars: FALSE

Time labels: FALSE

Arm: FALSE

Reference arm: FALSE

Covariates: FALSE

2.1.5 Sources

Id: auto

Arm: none

Improvement direction: no_reliable_mapping

Stable threshold: safe_zero_fallback

2.1.6 Analyses

Core: descriptives, missingness, change, variability, trajectories

Figures: FALSE

Primary mixed-effects model: FALSE

Diagnostic outliers: FALSE

Correlations across time points: FALSE

Arm tests: FALSE

2.2 Dataset overview

N rows: 48

N columns: 4

2.2.1 Column profile

Table 1: Column profile

variable	class	numeric	unique n	missing n	missing pct
subject_id	character	FALSE	48	0	0
score_t0	numeric	TRUE	48	0	0
score_t1	numeric	TRUE	48	0	0
score_t2	numeric	TRUE	48	0	0

2.3 Detected variables

2.3.1 Id candidates

Table 2: Id candidates (block 1 of 2)

variable	semantic score	score	observed n	unique n	missing n	duplicated n
subject_id	120	140	48	48	0	0

Table 3: Id candidates (block 2 of 2)

variable	semantic score	structurally valid
subject_id	120	TRUE

2.3.2 Longitudinal

Score: score_t0, score_t1, score_t2

2.3.3 Longitudinal map

Table 4: Longitudinal map

outcome	outcome key	variable	time label	time order	position	pattern	numeric
score	score	score_t0	t0	0	2	t	TRUE
score	score	score_t1	t1	1	3	t	TRUE
score	score	score_t2	t2	2	4	t	TRUE

3 Outcome: SCORE

3.1 Analysis settings

Alpha: 0.05

Confidence level: 0.95

Confidence percent: 95

P adjust method: holm

Improvement direction: unknown

Stable threshold: 0

Arm tests: FALSE

Strict id: TRUE

Non finite handling: Inf/-Inf treated as NA for analysis

Time vars: score_t0, score_t1, score_t2

Time labels: t0, t1, t2

3.2 Overview

Outcome: score

Outcome display: SCORE

Id: subject_id

N rows: 48

N patients: 48

N timepoints: 3

Timepoints: score_t0, score_t1, score_t2

Time labels: t0, t1, t2

Supplied time vars: TRUE

Missing ids: 0

Duplicated ids: 0

Complete profiles: 48

Complete profiles pct: 100

Non finite values: 0

Missing arm: 0

3.3 Descriptive statistics

Table 5: Descriptive statistics (block 1 of 6)

time	label	n	missing	missing pct	non finite	non finite pct	unavailable
score_t0	t0	48	0	0	0	0	0
score_t1	t1	48	0	0	0	0	0
score_t2	t2	48	0	0	0	0	0

Table 6: Descriptive statistics (block 2 of 6)

time	label	unavailable pct	mean	sd	variance
score_t0	t0	0	50.380	6.816	46.460
score_t1	t1	0	48.294	7.162	51.296
score_t2	t2	0	45.376	7.193	51.732

Table 7: Descriptive statistics (block 3 of 6)

time	label	se	ci lower	ci upper	median
score_t0	t0	0.984	48.401	52.359	49.41
score_t1	t1	1.034	46.214	50.373	47.71
score_t2	t2	1.038	43.287	47.464	44.13

Table 8: Descriptive statistics (block 4 of 6)

time	label	q1	q3	iqr	min
score_t0	t0	45.818	54.157	8.339	37.911
score_t1	t1	43.978	52.879	8.901	34.953
score_t2	t2	40.777	49.066	8.289	33.261

Table 9: Descriptive statistics (block 5 of 6)

time	label	max	cv percent	score mean	score sd
score_t0	t0	72.464	13.530	50.380	6.816
score_t1	t1	70.026	14.830	48.294	7.162
score_t2	t2	65.363	15.851	45.376	7.193

Table 10: Descriptive statistics (block 6 of 6)

time	label	score median	score ci lower	score ci upper
score_t0	t0	49.41	48.401	52.359
score_t1	t1	47.71	46.214	50.373
score_t2	t2	44.13	43.287	47.464

3.4 Missing data

3.4.1 By time point

Table 11: By time point (block 1 of 2)

time	label	missing n	missing pct	non finite n	non finite pct	unavailable n
score_t0	t0	0	0	0	0	0
score_t1	t1	0	0	0	0	0

time	label	missing n	missing pct	non finite n	non finite pct	unavailable n
score_t2	t2	0	0	0	0	0

Table 12: By time point (block 2 of 2)

time	label	unavailable pct
score_t0	t0	0
score_t1	t1	0
score_t2	t2	0

3.4.2 By participant

Table 13: By participant

patient	missing n	non finite n	unavailable n	unavailable pct
S001	0	0	0	0
S002	0	0	0	0
S003	0	0	0	0
S004	0	0	0	0
S005	0	0	0	0
S006	0	0	0	0
S007	0	0	0	0
S008	0	0	0	0
S009	0	0	0	0
S010	0	0	0	0
S011	0	0	0	0
S012	0	0	0	0
S013	0	0	0	0
S014	0	0	0	0
S015	0	0	0	0
S016	0	0	0	0
S017	0	0	0	0
S018	0	0	0	0
S019	0	0	0	0
S020	0	0	0	0
S021	0	0	0	0
S022	0	0	0	0
S023	0	0	0	0
S024	0	0	0	0
S025	0	0	0	0
S026	0	0	0	0
S027	0	0	0	0
S028	0	0	0	0
S029	0	0	0	0
S030	0	0	0	0
S031	0	0	0	0
S032	0	0	0	0
S033	0	0	0	0
S034	0	0	0	0
S035	0	0	0	0
S036	0	0	0	0
S037	0	0	0	0
S038	0	0	0	0
S039	0	0	0	0
S040	0	0	0	0
S041	0	0	0	0

patient	missing n	non finite n	unavailable n	unavailable pct
S042	0	0	0	0
S043	0	0	0	0
S044	0	0	0	0
S045	0	0	0	0
S046	0	0	0	0
S047	0	0	0	0
S048	0	0	0	0

3.5 Within-participant changes

Table 14: Within-participant changes (block 1 of 7)

from	to	from label	to label	n	mean from	mean to
score_t0	score_t1	t0	t1	48	50.380	48.294
score_t0	score_t2	t0	t2	48	50.380	45.376
score_t1	score_t2	t1	t2	48	48.294	45.376

Table 15: Within-participant changes (block 2 of 7)

from	to	mean change	sd change	se change	ci lower
score_t0	score_t1	-2.086	1.747	0.252	-2.593
score_t0	score_t2	-5.004	2.127	0.307	-5.622
score_t1	score_t2	-2.918	2.577	0.372	-3.666

Table 16: Within-participant changes (block 3 of 7)

from	to	ci upper	cohens dz	increased n	increased pct
score_t0	score_t1	-1.579	-1.194	4	8.333
score_t0	score_t2	-4.387	-2.353	0	0.000
score_t1	score_t2	-2.170	-1.133	5	10.417

Table 17: Within-participant changes (block 4 of 7)

from	to	decreased n	decreased pct	stable n	stable pct	improved n
score_t0	score_t1	44	91.667	0	0	–
score_t0	score_t2	48	100.000	0	0	–
score_t1	score_t2	43	89.583	0	0	–

Table 18: Within-participant changes (block 5 of 7)

from	to	improved pct	worsened n	worsened pct	paired t p	wilcoxon p
score_t0	score_t1	–	–	–	<0.001	<0.001
score_t0	score_t2	–	–	–	<0.001	<0.001
score_t1	score_t2	–	–	–	<0.001	<0.001

Table 19: Within-participant changes (block 6 of 7)

from	to	paired t p adj	wilcoxon p adj	score change	score mean from
score_t0	score_t1	<0.001	<0.001	-2.086	50.380

from	to	paired t p adj	wilcoxon p adj	score change	score mean from
score_t0	score_t2	<0.001	<0.001	-5.004	50.380
score_t1	score_t2	<0.001	<0.001	-2.918	48.294

Table 20: Within-participant changes (block 7 of 7)

from	to	score mean to
score_t0	score_t1	48.294
score_t0	score_t2	45.376
score_t1	score_t2	45.376

3.6 Variability and ICC

Table 21: Variability and ICC (block 1 of 4)

outcome	grand mean	between subject sd	within subject sd	ICC anova complete profiles	ICC model
score	48.016	6.946	2.408	0.84	–

Table 22: Variability and ICC (block 2 of 4)

outcome	grand mean	ms between	ms within	df between	df within
score	48.016	144.751	8.637	47	96

Table 23: Variability and ICC (block 3 of 4)

outcome	grand mean	ICC	score grand mean	score between subject sd
score	48.016	0.84	48.016	6.946

Table 24: Variability and ICC (block 4 of 4)

outcome	grand mean	score within subject sd
score	48.016	2.408

3.7 Individual trajectories

Table 25: Individual trajectories (block 1 of 3)

patient	outcome	arm	baseline	final	absolute change	relative change percent
S001	score	–	50.562	40.924	-9.638	-19.062
S002	score	–	50.382	44.150	-6.232	-12.369
S003	score	–	45.186	39.227	-5.959	-13.188
S004	score	–	54.964	54.907	-0.057	-0.104
S005	score	–	40.388	36.304	-4.084	-10.111
S006	score	–	49.283	42.728	-6.554	-13.300
S007	score	–	44.439	39.632	-4.807	-10.818
S008	score	–	44.376	36.162	-8.214	-18.510
S009	score	–	53.031	48.245	-4.786	-9.025
S010	score	–	60.953	58.055	-2.898	-4.754

patient	outcome	arm	baseline	final	absolute change	relative change percent
S011	score	—	58.478	52.623	-5.855	-10.012
S012	score	—	56.951	50.635	-6.316	-11.089
S013	score	—	47.385	40.318	-7.067	-14.915
S014	score	—	56.570	55.080	-1.491	-2.635
S015	score	—	63.154	56.425	-6.729	-10.655
S016	score	—	40.681	34.373	-6.308	-15.506
S017	score	—	45.926	42.454	-3.472	-7.560
S018	score	—	49.882	43.070	-6.813	-13.657
S019	score	—	38.900	33.261	-5.638	-14.495
S020	score	—	47.851	44.702	-3.149	-6.581
S021	score	—	62.633	58.064	-4.569	-7.295
S022	score	—	55.693	47.658	-8.035	-14.427
S023	score	—	50.717	46.402	-4.316	-8.509
S024	score	—	44.029	35.139	-8.891	-20.193
S025	score	—	48.042	46.144	-1.898	-3.950
S026	score	—	54.104	49.757	-4.347	-8.035
S027	score	—	49.525	42.898	-6.627	-13.380
S028	score	—	61.673	57.879	-3.794	-6.152
S029	score	—	52.382	48.835	-3.547	-6.771
S030	score	—	49.296	46.974	-2.322	-4.710
S031	score	—	46.961	41.288	-5.673	-12.081
S032	score	—	55.058	53.432	-1.626	-2.953
S033	score	—	49.275	45.127	-4.147	-8.416
S034	score	—	45.495	42.182	-3.313	-7.281
S035	score	—	54.317	50.356	-3.961	-7.293
S036	score	—	47.753	41.825	-5.929	-12.415
S037	score	—	51.344	48.299	-3.045	-5.930
S038	score	—	41.518	34.791	-6.727	-16.202
S039	score	—	44.823	40.336	-4.487	-10.010
S040	score	—	47.796	41.177	-6.619	-13.848
S041	score	—	46.045	38.981	-7.064	-15.341
S042	score	—	72.464	65.363	-7.101	-9.799
S043	score	—	48.672	44.111	-4.561	-9.371
S044	score	—	53.525	46.442	-7.083	-13.233
S045	score	—	52.396	48.750	-3.646	-6.958
S046	score	—	49.983	43.322	-6.661	-13.326
S047	score	—	45.463	42.275	-3.188	-7.012
S048	score	—	37.911	36.949	-0.962	-2.538

Table 26: Individual trajectories (block 2 of 3)

patient	outcome	direction	clinical direction	score baseline	score final
S001	score	Decrease	—	50.562	40.924
S002	score	Decrease	—	50.382	44.150
S003	score	Decrease	—	45.186	39.227
S004	score	Decrease	—	54.964	54.907
S005	score	Decrease	—	40.388	36.304
S006	score	Decrease	—	49.283	42.728
S007	score	Decrease	—	44.439	39.632
S008	score	Decrease	—	44.376	36.162
S009	score	Decrease	—	53.031	48.245
S010	score	Decrease	—	60.953	58.055
S011	score	Decrease	—	58.478	52.623
S012	score	Decrease	—	56.951	50.635
S013	score	Decrease	—	47.385	40.318

patient	outcome	direction	clinical direction	score baseline	score final
S014	score	Decrease	–	56.570	55.080
S015	score	Decrease	–	63.154	56.425
S016	score	Decrease	–	40.681	34.373
S017	score	Decrease	–	45.926	42.454
S018	score	Decrease	–	49.882	43.070
S019	score	Decrease	–	38.900	33.261
S020	score	Decrease	–	47.851	44.702
S021	score	Decrease	–	62.633	58.064
S022	score	Decrease	–	55.693	47.658
S023	score	Decrease	–	50.717	46.402
S024	score	Decrease	–	44.029	35.139
S025	score	Decrease	–	48.042	46.144
S026	score	Decrease	–	54.104	49.757
S027	score	Decrease	–	49.525	42.898
S028	score	Decrease	–	61.673	57.879
S029	score	Decrease	–	52.382	48.835
S030	score	Decrease	–	49.296	46.974
S031	score	Decrease	–	46.961	41.288
S032	score	Decrease	–	55.058	53.432
S033	score	Decrease	–	49.275	45.127
S034	score	Decrease	–	45.495	42.182
S035	score	Decrease	–	54.317	50.356
S036	score	Decrease	–	47.753	41.825
S037	score	Decrease	–	51.344	48.299
S038	score	Decrease	–	41.518	34.791
S039	score	Decrease	–	44.823	40.336
S040	score	Decrease	–	47.796	41.177
S041	score	Decrease	–	46.045	38.981
S042	score	Decrease	–	72.464	65.363
S043	score	Decrease	–	48.672	44.111
S044	score	Decrease	–	53.525	46.442
S045	score	Decrease	–	52.396	48.750
S046	score	Decrease	–	49.983	43.322
S047	score	Decrease	–	45.463	42.275
S048	score	Decrease	–	37.911	36.949

Table 27: Individual trajectories (block 3 of 3)

patient	outcome	score change	score relative change percent
S001	score	-9.638	-19.062
S002	score	-6.232	-12.369
S003	score	-5.959	-13.188
S004	score	-0.057	-0.104
S005	score	-4.084	-10.111
S006	score	-6.554	-13.300
S007	score	-4.807	-10.818
S008	score	-8.214	-18.510
S009	score	-4.786	-9.025
S010	score	-2.898	-4.754
S011	score	-5.855	-10.012
S012	score	-6.316	-11.089
S013	score	-7.067	-14.915
S014	score	-1.491	-2.635
S015	score	-6.729	-10.655
S016	score	-6.308	-15.506
S017	score	-3.472	-7.560

patient	outcome	score change	score relative change percent
S018	score	-6.813	-13.657
S019	score	-5.638	-14.495
S020	score	-3.149	-6.581
S021	score	-4.569	-7.295
S022	score	-8.035	-14.427
S023	score	-4.316	-8.509
S024	score	-8.891	-20.193
S025	score	-1.898	-3.950
S026	score	-4.347	-8.035
S027	score	-6.627	-13.380
S028	score	-3.794	-6.152
S029	score	-3.547	-6.771
S030	score	-2.322	-4.710
S031	score	-5.673	-12.081
S032	score	-1.626	-2.953
S033	score	-4.147	-8.416
S034	score	-3.313	-7.281
S035	score	-3.961	-7.293
S036	score	-5.929	-12.415
S037	score	-3.045	-5.930
S038	score	-6.727	-16.202
S039	score	-4.487	-10.010
S040	score	-6.619	-13.848
S041	score	-7.064	-15.341
S042	score	-7.101	-9.799
S043	score	-4.561	-9.371
S044	score	-7.083	-13.233
S045	score	-3.646	-6.958
S046	score	-6.661	-13.326
S047	score	-3.188	-7.012
S048	score	-0.962	-2.538

3.8 Advanced longitudinal tests

3.8.1 Estimated marginal means and contrasts

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.8.2 Repeated-measures ANOVA

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.8.3 Friedman test

Object

```
Friedman rank sum test

data:  as.matrix(complete)
Friedman chi-squared = 78.875, df = 2, p-value < 2.2e-16
```

Test

```
Friedman rank sum test

data:  as.matrix(complete)
Friedman chi-squared = 78.875, df = 2, p-value < 2.2e-16
```

Model summary

Table 28: Model summary

statistic	df	p raw	kendalls w	n	n timepoints
78.875	2	<0.001	0.822	48	3

Posthoc

Tests

Score t0 to score t1

Wilcoxon signed rank test with continuity correction

data: y and x

V = 57, p-value = 5.295e-08

alternative hypothesis: true location shift is not equal to 0

Score t0 to score t2

Wilcoxon signed rank test with continuity correction

data: y and x

V = 0, p-value = 1.684e-09

alternative hypothesis: true location shift is not equal to 0

Score t1 to score t2

Wilcoxon signed rank test with continuity correction

data: y and x

V = 55, p-value = 4.718e-08

alternative hypothesis: true location shift is not equal to 0

Tidy

Table 29: Tidy (block 1 of 2)

from	to	from label	to label	n	statistic	p raw	rank biserial
score_t0	score_t1	t0	t1	48	57	<0.001	-0.903
score_t0	score_t2	t0	t2	48	0	<0.001	-1.000
score_t1	score_t2	t1	t2	48	55	<0.001	-0.906

Table 30: Tidy (block 2 of 2)

from	to	p primary	primary method	p bonferroni	p holm	p bh	p by
score_t0	score_t1	<0.001	holm	<0.001	<0.001	<0.001	<0.001
score_t0	score_t2	<0.001	holm	<0.001	<0.001	<0.001	<0.001
score_t1	score_t2	<0.001	holm	<0.001	<0.001	<0.001	<0.001

3.9 Advanced longitudinal models

3.9.1 Random-slope mixed-effects model

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.9.2 NLME models

Compound-symmetry correlation

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

AR(1) correlation

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.9.3 GEE models

Independent working correlation

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

Exchangeable working correlation

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

AR(1) correlation

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.10 Robust inference

3.10.1 Cluster-robust CR2 inference

Reason skipped: Advanced model-based analyses were disabled by model = FALSE.

3.11 Effect sizes

3.11.1 Paired cohens dz

Table 31: Paired cohens dz

from	to	from label	to label	n	cohens dz
score_t0	score_t1	t0	t1	48	-1.194
score_t0	score_t2	t0	t2	48	-2.353
score_t1	score_t2	t1	t2	48	-1.133

3.11.2 Paired rank biserial

Table 32: Paired rank biserial

from	to	from label	to label	n	rank biserial
score_t0	score_t1	t0	t1	48	-0.903
score_t0	score_t2	t0	t2	48	-1.000
score_t1	score_t2	t1	t2	48	-0.906

3.11.3 Friedman test

Kendalls w: 0.8216145833333333

3.11.4 Mixed models

Random intercept

Reason skipped: The mixed model was not available.

Random-slope mixed-effects model

Reason skipped: The mixed model was not available.

3.12 Multiplicity families

Primary method: holm

Note: P-values are adjusted within scientifically distinct contrast families.

3.12.1 Friedman posthoc

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Table

Table 33: Table (block 1 of 2)

from	to	from label	to label	n	statistic	p raw	rank biserial
score_t0	score_t1	t0	t1	48	57	<0.001	-0.903
score_t0	score_t2	t0	t2	48	0	<0.001	-1.000
score_t1	score_t2	t1	t2	48	55	<0.001	-0.906

Table 34: Table (block 2 of 2)

from	to	p primary	primary method	p bonferroni	p holm	p bh	p by
score_t0	score_t1	<0.001	holm	<0.001	<0.001	<0.001	<0.001
score_t0	score_t2	<0.001	holm	<0.001	<0.001	<0.001	<0.001
score_t1	score_t2	<0.001	holm	<0.001	<0.001	<0.001	<0.001

3.13 Sensitivity analyses

Table 35: Sensitivity analyses (block 1 of 3)

question	method	statistic	df	p raw	effect size	effect size type	n
TIME	Friedman test	78.875	2	<0.001	0.822	Kendall's W	48

Table 36: Sensitivity analyses (block 2 of 3)

question	method	object path
TIME	Friedman test	\$advanced_tests\$friedman\$test

Table 37: Sensitivity analyses (block 3 of 3)

question	method	note
TIME	Friedman test	Rank-based complete-profile global test.

Interpretation: Use this table to compare sensitivity across methods. It is not a majority vote and does not designate an effect as true based on the number of small p-values.

3.14 Diagnostics and adaptations

3.14.1 Automatic adaptations

Available by time: 48, 48, 48

Variable timepoints: 3

4 Descriptive summary

Table 38: Differences between observed means

outcome	mean first visit	mean last visit	descriptive difference
score	50.38	45.376	-5.004